

A decorative graphic on the left side of the slide. It features several vertical stripes of varying shades of blue and grey. Overlaid on these stripes are several circles of different sizes, also in shades of blue and grey, arranged in a cluster.

# **RELATIONAL DATABASE MODEL AND CONSTRAINTS**

# FLAT DATABASE

- Issues ?

- Difficult to search
- Difficult to generate reports
- Difficult to sort data based on particular field



# RELATIONAL DATABASE

- Relational databases use **tables** to store information.

| LName  | FName | City     | Age | Salary |
|--------|-------|----------|-----|--------|
| Smith  | John  | New York | 35  | \$280  |
| Doe    | Jane  | Boston   | 28  | \$325  |
| Brown  | Scott | New York | 41  | \$265  |
| Howard | Shemp | New York | 48  | \$359  |
| Taylor | Tom   | London   | 22  | \$250  |

- *Issue with this table ?*
- **Primary Key:** uniquely identifies each record in the table

# RELATIONAL DATABASE

- “Relational” part ->
  - how multiple tables relate to each other,
  - allowing a user to extract information from a combination of tables

| LName  | FName | City | Age | Salary |
|--------|-------|------|-----|--------|
| Smith  | John  | 3    | 35  | \$280  |
| Doe    | Jane  | 1    | 28  | \$325  |
| Brown  | Scott | 3    | 41  | \$265  |
| Howard | Shemp | 4    | 48  | \$359  |
| Taylor | Tom   | 2    | 22  | \$250  |

| City # | City Name   |
|--------|-------------|
| 1      | Boston      |
| 2      | London      |
| 3      | New York    |
| 4      | Los Angeles |



# RELATIONAL DATABASE VS SPREADSHEET

- Relational database is a **big spreadsheet**
  - that several people can update simultaneously
- Each *table* in the database is one spreadsheet
- **Difference:**
  - RD is more restrictive than a spreadsheet
    - All the data in one column must be of the same type, e.g., integer, decimal, character string, or date.
  - Rows in an RD are not ordered
  - RD allow user to enforce constraints
- Example: Mailing List



# FORMAL DEFINITIONS

- **Schema** of a Relation:  $R (A_1, A_2, \dots A_n)$
- Example:  
STUDENT(Name, Ssn, Phone, Address, Age, Gpa)
- Each attribute has a **domain** or a set of valid values.
- Example:
  - Ssn: set of valid 9 digit social security numbers.
  - Phone: the set of 10 digit phone numbers valid in the United States.



# FORMAL DEFINITIONS

- Relation State  $r(\mathbf{R})$  is set of  $n$ -tuples
- Each  **$n$ -tuple**  $t$  is an ordered list of  $n$  values

$$t = \langle v_1, v_2, \dots, v_n \rangle$$

where each value  $v_i$ ,  $1 \leq i \leq n$ , is an element of  $\text{dom}(A_i)$  or is NULL .

The diagram illustrates the formal definition of a relation state. It shows a table with 7 columns and 6 rows. The columns are labeled 'Name', 'Ssn', 'Home\_phone', 'Address', 'Office\_phone', 'Age', and 'Gpa'. The rows contain data for five students: Benjamin Bayer, Chung-cha Kim, Dick Davidson, Rohan Panchal, and Barbara Benson. The last row is empty. Arrows point from 'Relation Name' to 'STUDENT' and from 'Attributes' to each column header. A bracket on the left labeled 'Tuples' points to each row of data.

|  | Name           | Ssn         | Home_phone    | Address              | Office_phone  | Age | Gpa  |
|--|----------------|-------------|---------------|----------------------|---------------|-----|------|
|  | Benjamin Bayer | 305-61-2435 | (817)373-1616 | 2918 Bluebonnet Lane | NULL          | 19  | 3.21 |
|  | Chung-cha Kim  | 381-62-1245 | (817)375-4409 | 125 Kirby Road       | NULL          | 18  | 2.89 |
|  | Dick Davidson  | 422-11-2320 | NULL          | 3452 Elgin Road      | (817)749-1253 | 25  | 3.53 |
|  | Rohan Panchal  | 489-22-1100 | (817)376-9821 | 265 Lark Lane        | (817)749-6492 | 28  | 3.93 |
|  | Barbara Benson | 533-69-1238 | (817)839-8461 | 7384 Fontana Lane    | NULL          | 19  | 3.25 |
|  |                |             |               |                      |               |     |      |

# FORMAL DEFINITIONS

- A relation  $r(R)$  is a **mathematical relation** of degree  $n$  on the domains  $\text{dom}(A_1), \dots, \text{dom}(A_n)$ .

- $r(R)$  is a **subset of Cartesian product** of the domains that define  $R$

$$r(R) \subseteq (\text{dom}(A_1) \times \text{dom}(A_2) \times \dots \times \text{dom}(A_n))$$

- Example:

- Let  $S1 = \{0,1\}$  and  $S2 = \{a,b,c\}$
- $r(R) \subset S1 \times S2$
- $r(R) = \{ \langle 0,a \rangle, \langle 0,b \rangle, \langle 1,c \rangle \}$  is one possible “state”





# DEFINITIONS USING SET THEORY

- The total no of tuples in the Cartesian product is

$$|\text{dom}(A1)| \times |\text{dom}(A2)| \times \dots \times |\text{dom}(A_n)|$$

- This shows the total no of possible tuples that can ever exist in any relation state  $r(R)$ .



| <u>Informal Terms</u> |  | <u>Formal Terms</u>  |
|-----------------------|--|----------------------|
|                       |  |                      |
| Table                 |  | Relation             |
| Column                |  | Attribute/Domain     |
| Row                   |  | Tuple                |
| Values in a column    |  | Domain               |
| Table Definition      |  | Schema of a Relation |
| Populated Table       |  | Extension            |



# CHARACTERISTICS OF RELATIONS

- Ordering of tuples in a relation  $r(R)$ :

**STUDENT**

| Name           | Ssn         | Home_phone    | Address              | Office_phone  | Age | Gpa  |
|----------------|-------------|---------------|----------------------|---------------|-----|------|
| Dick Davidson  | 422-11-2320 | NULL          | 3452 Elgin Road      | (817)749-1253 | 25  | 3.53 |
| Barbara Benson | 533-69-1238 | (817)839-8461 | 7384 Fontana Lane    | NULL          | 19  | 3.25 |
| Rohan Panchal  | 489-22-1100 | (817)376-9821 | 265 Lark Lane        | (817)749-6492 | 28  | 3.93 |
| Chung-cha Kim  | 381-62-1245 | (817)375-4409 | 125 Kirby Road       | NULL          | 18  | 2.89 |
| Benjamin Bayer | 305-61-2435 | (817)373-1616 | 2918 Bluebonnet Lane | NULL          | 19  | 3.21 |



# CHARACTERISTICS OF RELATIONS

- Ordering of attributes in a relation schema R
  - We define *n*-tuple as an *ordered list* of *n* values
  - Thus, attributes in  $R(A_1, A_2, \dots, A_n)$  and the values in  $t = \langle v_1, v_2, \dots, v_n \rangle$  are considered to be ordered.

**STUDENT**

| Name           | Ssn         | Home_phone    | Address              | Office_phone  | Age | Gpa  |
|----------------|-------------|---------------|----------------------|---------------|-----|------|
| Dick Davidson  | 422-11-2320 | NULL          | 3452 Elgin Road      | (817)749-1253 | 25  | 3.53 |
| Barbara Benson | 533-69-1238 | (817)839-8461 | 7384 Fontana Lane    | NULL          | 19  | 3.25 |
| Rohan Panchal  | 489-22-1100 | (817)376-9821 | 265 Lark Lane        | (817)749-6492 | 28  | 3.93 |
| Chung-cha Kim  | 381-62-1245 | (817)375-4409 | 125 Kirby Road       | NULL          | 18  | 2.89 |
| Benjamin Bayer | 305-61-2435 | (817)373-1616 | 2918 Bluebonnet Lane | NULL          | 19  | 3.21 |

# CHARACTERISTICS OF RELATIONS

- An alternative definition of a relation can eliminate the need for ordering in tuple
  - Define **tuple** as a set of ( $\langle \text{attribute} \rangle$ ,  $\langle \text{value} \rangle$ ) pairs
  - Each pair gives the value of the mapping from an attribute  $A_i$  to a value  $v_i$  from  $\text{dom}(A_i)$ .

Two identical tuples when the order of attributes and values is not part of relation definition.

$t = \langle (\text{Name}, \text{Dick Davidson}), (\text{Ssn}, 422-11-2320), (\text{Home\_phone}, \text{NULL}), (\text{Address}, 3452 \text{ Elgin Road}), (\text{Office\_phone}, (817)749-1253), (\text{Age}, 25), (\text{Gpa}, 3.53) \rangle$

$t = \langle (\text{Address}, 3452 \text{ Elgin Road}), (\text{Name}, \text{Dick Davidson}), (\text{Ssn}, 422-11-2320), (\text{Age}, 25), (\text{Office\_phone}, (817)749-1253), (\text{Gpa}, 3.53), (\text{Home\_phone}, \text{NULL}) \rangle$



# CHARACTERISTICS OF RELATIONS

- Values in a tuple:
  - Atomic (indivisible).
  - Null value
- Domains of all attributes should be indivisible
  - Integers and strings,
  - Record structures: set, list, array ... ?



# RELATIONS

- Relations represent facts about *entities and relationships*.
- For example, consider a relation schema  
MAJORS(Student\_ssn, Department\_code)
  - It asserts that students major in academic disciplines.
  - A tuple in this relation relates a student to his or her major discipline.



# WHY WE NEED MULTIPLE TABLES

- Why not put all attributes in one relation ?

| EMPLOYEE |       |       |            |       |         |     |        |          |     |
|----------|-------|-------|------------|-------|---------|-----|--------|----------|-----|
| FNAME    | MINIT | LNAME | <u>SSN</u> | BDATE | ADDRESS | SEX | SALARY | SUPERSSN | DNO |

| DEPARTMENT |                |        |              |
|------------|----------------|--------|--------------|
| DNAME      | <u>DNUMBER</u> | MGRSSN | MGRSTARTDATE |

| DEPT_LOCATIONS |                  |
|----------------|------------------|
| <u>DNUMBER</u> | <u>DLOCATION</u> |

- Duplication
- Primary key
- Cannot insert value for employee with no assigned department ... Null value in PK





# RELATIONAL DATABASE CONSTRAINTS

- Generally, there are many constraints on the values in a database state.
- Constraints are derived from the rules in the mini-world



Up-till now we have seen constraints

- **No duplicate tuples** (rows)
- **Unique id** (primary key )
- **Atomic value** for domain of an attribute



# CATEGORIES OF CONSTRAINTS

- **Model-based constraints or implicit constraints.**
  - These are inherent in data model, like no duplicate rows in RDBMs table.
- **Schema-based or explicit constraints.**
  - Can be expressed directly in the schema using DDL
- **Application based or semantic constraints or business rules.**
  - Can't be expressed directly in the schema
  - Must be enforced by the application programs



# SCHEMA-BASED CONSTRAINTS

- **Domain** constraints
- **Key** constraints
- **Entity integrity** constraints
- **Referential integrity** constraints



# KEY CONSTRAINT

- Relation is a **set** of tuples. Thus, all tuples must be distinct as all elements in the set are distinct

- **Example 1: Student Relation**

*Student(Name, SSN, Address, Phone, Major, Age, Gpa )*

- **KEY** is SSN, as it uniquely identify the student.

- **Example 2: Department Relation**

*Department(Dname, Dnumber, MgrSSN, Mgrstrdate)*

- **KEY** is Dnumber
- Candidate Key Dname



# EXAMPLE OF A UNIVERSITY DATABASE

## COURSE

| Course_name               | Course_number | Credit_hours | Department |
|---------------------------|---------------|--------------|------------|
| Intro to Computer Science | CS1310        | 4            | CS         |
| Data Structures           | CS3320        | 4            | CS         |
| Discrete Mathematics      | MATH2410      | 3            | MATH       |
| Database                  | CS3380        | 3            | CS         |

## SECTION

| Section_identifier | Course_number | Semester | Year | Instructor |
|--------------------|---------------|----------|------|------------|
| 85                 | MATH2410      | Fall     | 04   | King       |
| 92                 | CS1310        | Fall     | 04   | Anderson   |
| 102                | CS3320        | Spring   | 05   | Knuth      |
| 112                | MATH2410      | Fall     | 05   | Chang      |
| 119                | CS1310        | Fall     | 05   | Anderson   |
| 135                | CS3380        | Fall     | 05   | Stone      |

## GRADE\_REPORT

| Student_number | Section_identifier | Grade |
|----------------|--------------------|-------|
| 17             | 112                | B     |
| 17             | 119                | C     |
| 8              | 85                 | A     |
| 8              | 92                 | A     |
| 8              | 102                | B     |
| 8              | 135                | A     |

## PREREQUISITE

| Course_number | Prerequisite_number |
|---------------|---------------------|
| CS3380        | CS3320              |
| CS3380        | MATH2410            |
| CS3320        | CS1310              |

**Figure 1.2**  
A database that stores  
student and course  
information.



# KEY CONSTRAINT

- **Superkey:**

- For any distinct tuples  $t_1$  &  $t_2$  in  $r(R)$ ,  
 $t_1[SK] \neq t_2[SK]$ .

- **Key:**

- A "minimal" superkey;
- Key is a superkey from which we cannot remove any attribute and still have the uniqueness constraint hold.

- **Example:**

- *Student(Name, SSN, Address, Phone, Major, Age, Gpa )*

- Consider set of attributes {Name, SSN, Age},
  - This set uniquely identify the student. This is **SUPERKEY**
  - **KEY** is SSN, minimal set that uniquely identify the student.



# KEY CONSTRAINT

- Consider Property Schema

|                     |             |      |      |       |          |
|---------------------|-------------|------|------|-------|----------|
| <u>Property_id#</u> | County_name | Lot# | Area | Price | Tax_rate |
|---------------------|-------------|------|------|-------|----------|

- Key is Property\_id#
- Candidate Key (Country\_name, Lot#)
  - This is a minimal superkey, as we cannot remove any attribute and still have a uniqueness constraint hold.





# KEY CONSTRAINT

## ○ Example 2: CAR relation schema

**CAR(LicenseNo, Engine\_SerialNo, Make, Model, Year)**

## ○ Key

- Key1 = {LicenseNo}
- Key2 = {Engine\_SerialNo},
- These are also superkeys.

## ○ Superkey: {Engine\_SerialNo, Make}

## ○ Candidate keys: If a relation has more than one key, we call it **candidate keys**



**Figure 7.4** The CAR relation with two candidate keys:  
LicenseNumber and EngineSerialNumber.

| CAR | <u>LicenseNumber</u> | EngineSerialNumber | Make       | Model   | Year |
|-----|----------------------|--------------------|------------|---------|------|
|     | Texas ABC-739        | A69352             | Ford       | Mustang | 96   |
|     | Florida TVP-347      | B43696             | Oldsmobile | Cutlass | 99   |
|     | New York MPO-22      | X83554             | Oldsmobile | Delta   | 95   |
|     | California 432-TFY   | C43742             | Mercedes   | 190-D   | 93   |
|     | California RSK-629   | Y82935             | Toyota     | Camry   | 98   |
|     | Texas RSK-629        | U028365            | Jaguar     | XJS     | 98   |



### EMPLOYEE

|       |       |       |            |       |         |     |        |          |     |
|-------|-------|-------|------------|-------|---------|-----|--------|----------|-----|
| FNAME | MINIT | LNAME | <u>SSN</u> | BDATE | ADDRESS | SEX | SALARY | SUPERSSN | DNO |
|-------|-------|-------|------------|-------|---------|-----|--------|----------|-----|

### DEPARTMENT

|       |                |        |              |
|-------|----------------|--------|--------------|
| DNAME | <u>DNUMBER</u> | MGRSSN | MGRSTARTDATE |
|-------|----------------|--------|--------------|

### DEPT\_LOCATIONS

|                |                  |
|----------------|------------------|
| <u>DNUMBER</u> | <u>DLOCATION</u> |
|----------------|------------------|

### PROJECT

|       |                |           |      |
|-------|----------------|-----------|------|
| PNAME | <u>PNUMBER</u> | PLOCATION | DNUM |
|-------|----------------|-----------|------|

### WORKS\_ON

|             |            |       |
|-------------|------------|-------|
| <u>ESSN</u> | <u>PNO</u> | HOURS |
|-------------|------------|-------|

### DEPENDENT

|             |                       |     |       |              |
|-------------|-----------------------|-----|-------|--------------|
| <u>ESSN</u> | <u>DEPENDENT_NAME</u> | SEX | BDATE | RELATIONSHIP |
|-------------|-----------------------|-----|-------|--------------|

# INTEGRITY CONSTRAINTS

- Integrity constraints are specified on DB schema
- They are expected to hold on every valid database state of the schema.
- Integrity constraints ensure accuracy and consistency of data in a relational database.



# ENTITY INTEGRITY

- Primary key is used to *identify* the individual tuples so it cannot have null value

Formally,

- Schema  $S$  consist of many relations  $S = \{R_1, R_2, \dots, R_n\}$  and State  $DB\{r_1, r_2, \dots, r_n\}$  where each  $r_i$  is the state of  $R_i$ .
- Primary key attributes* of each relation schema  $R$  in  $S$  cannot have null values in any tuple of  $r(R)$ .

$t[PK] \neq \text{null}$  for any tuple  $t$  in  $r(R)$



**EMPLOYEE**

| Fname    | Minit | Lname   | Ssn       | Bdate      | Address                  | Sex | Salary | Super_ssn | Dno |
|----------|-------|---------|-----------|------------|--------------------------|-----|--------|-----------|-----|
| John     | B     | Smith   | 123456789 | 1965-01-09 | 731 Fondren, Houston, TX | M   | 30000  | 333445555 | 5   |
| Franklin | T     | Wong    | 333445555 | 1955-12-08 | 638 Voss, Houston, TX    | M   | 40000  | 888665555 | 5   |
| Alicia   | J     | Zelaya  | 999887777 | 1968-01-19 | 3321 Castle, Spring, TX  | F   | 25000  | 987654321 | 4   |
| Jennifer | S     | Wallace | 987654321 | 1941-06-20 | 291 Berry, Bellaire, TX  | F   | 43000  | 888665555 | 4   |
| Ramesh   | K     | Narayan | 666884444 | 1962-09-15 | 975 Fire Oak, Humble, TX | M   | 38000  | 333445555 | 5   |
| Joyce    | A     | English | 453453453 | 1972-07-31 | 5631 Rice, Houston, TX   | F   | 25000  | 333445555 | 5   |
| Ahmad    | V     | Jabbar  | 987987987 | 1969-03-29 | 980 Dallas, Houston, TX  | M   | 25000  | 987654321 | 4   |
| James    | E     | Borg    | 888665555 | 1937-11-10 | 450 Stone, Houston, TX   | M   | 55000  | NULL      | 1   |

**DEPARTMENT**

| Dname          | Dnumber | Mgr_ssn   | Mgr_start_date |
|----------------|---------|-----------|----------------|
| Research       | 5       | 333445555 | 1988-05-22     |
| Administration | 4       | 987654321 | 1995-01-01     |
| Headquarters   | 1       | 888665555 | 1981-06-19     |

**WORKS\_ON**

| Essn      | Pno | Hours |
|-----------|-----|-------|
| 123456789 | 1   | 32.5  |
| 123456789 | 2   | 7.5   |
| 666884444 | 3   | 40.0  |
| 453453453 | 1   | 20.0  |
| 453453453 | 2   | 20.0  |
| 333445555 | 2   | 10.0  |
| 333445555 | 3   | 10.0  |
| 333445555 | 10  | 10.0  |
| 333445555 | 20  | 10.0  |
| 999887777 | 30  | 30.0  |
| 999887777 | 10  | 10.0  |
| 987987987 | 10  | 35.0  |
| 987987987 | 30  | 5.0   |
| 987654321 | 30  | 20.0  |
| 987654321 | 20  | 15.0  |
| 888665555 | 20  | NULL  |

**PROJECT**

| Pname           | Pnumber | Plocation | Dnum |
|-----------------|---------|-----------|------|
| ProductX        | 1       | Bellaire  | 5    |
| ProductY        | 2       | Sugarland | 5    |
| ProductZ        | 3       | Houston   | 5    |
| Computerization | 10      | Stafford  | 4    |
| Reorganization  | 20      | Houston   | 1    |
| Newbenefits     | 30      | Stafford  | 4    |

**DEPENDENT**

| Essn      | Dependent_name | Sex | Bdate      | Relationship |
|-----------|----------------|-----|------------|--------------|
| 333445555 | Alice          | F   | 1986-04-05 | Daughter     |
| 333445555 | Theodore       | M   | 1983-10-25 | Son          |
| 333445555 | Joy            | F   | 1958-05-03 | Spouse       |
| 987654321 | Abner          | M   | 1942-02-28 | Spouse       |
| 123456789 | Michael        | M   | 1988-01-04 | Son          |
| 123456789 | Alice          | F   | 1988-12-30 | Daughter     |
| 123456789 | Elizabeth      | F   | 1967-05-05 | Spouse       |

**DEPT\_LOCATIONS**

| Dnumber | Dlocation |
|---------|-----------|
| 1       | Houston   |
| 4       | Stafford  |
| 5       | Bellaire  |
| 5       | Sugarland |
| 5       | Houston   |



# REFERENTIAL INTEGRITY

- *A tuple in one relation that refers to another relation must refer to an existing tuple in that relation.*
- This is specified to maintain consistency among tuples in the two relations.



# FOREIGN KEY (FK)

- A foreign key is a field in a table that matches the primary key column of another table.
- Set of attributes **FK** in relation  $R_1$  is called foreign key of  $R_1$  that reference  $R_2$  if
  - Attributes in FK in  $R_1$  has **same domain** as the attributes in PK of  $R_2$
  - **Value of FK** must be an existing PK value in  $R_2$  or null.

DEPARTMENT

| Dname          | <u>Dnumber</u> | Mgr_ssn   | Mgr_start_date |
|----------------|----------------|-----------|----------------|
| Research       | 5              | 333445555 | 1988-05-22     |
| Administration | 4              | 987654321 | 1995-01-01     |
| Headquarters   | 1              | 888665555 | 1981-06-19     |

DEPT\_LOCATIONS

| <u>Dnumber</u> | <u>Dlocation</u> |
|----------------|------------------|
| 1              | Houston          |
| 4              | Stafford         |
| 5              | Bellaire         |
| 5              | Sugarland        |
| 5              | Houston          |





# FOREIGN KEY (FK)

- Tuple  $t_1$  in  $R_1$  is said to reference a tuple  $t_2$  in  $R_2$  if  $t_1[FK] = t_2[PK]$ .

$R_1$  is referencing relation and  $R_2$  is reference relation

EMPLOYEE

| Fname    | Minit | Lname   | <u>Ssn</u> | Bdate      | Address                  | Sex | Salary | Super_ssn | Dno |
|----------|-------|---------|------------|------------|--------------------------|-----|--------|-----------|-----|
| John     | B     | Smith   | 123456789  | 1965-01-09 | 731 Fondren, Houston, TX | M   | 30000  | 333445555 | 5   |
| Franklin | T     | Wong    | 333445555  | 1955-12-08 | 638 Voss, Houston, TX    | M   | 40000  | 888665555 | 5   |
| Alicia   | J     | Zelaya  | 999887777  | 1968-01-19 | 3321 Castle, Spring, TX  | F   | 25000  | 987654321 | 4   |
| Jennifer | S     | Wallace | 987654321  | 1941-06-20 | 291 Berry, Bellaire, TX  | F   | 43000  | 888665555 | 4   |
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| Joyce    | A     | English | 453453453  | 1972-07-31 | 5631 Rice, Houston, TX   | F   | 25000  | 333445555 | 5   |
| Ahmad    | V     | Jabbar  | 987987987  | 1969-03-29 | 980 Dallas, Houston, TX  | M   | 25000  | 987654321 | 4   |
| James    | E     | Borg    | 888665555  | 1937-11-10 | 450 Stone, Houston, TX   | M   | 55000  | NULL      | 1   |

DEPARTMENT

| Dname          | <u>Dnumber</u> | Mgr_ssn   | Mgr_start_date |
|----------------|----------------|-----------|----------------|
| Research       | 5              | 333445555 | 1988-05-22     |
| Administration | 4              | 987654321 | 1995-01-01     |
| Headquarters   | 1              | 888665555 | 1981-06-19     |

DEPT\_LOCATIONS

| <u>Dnumber</u> | <u>Dlocation</u> |
|----------------|------------------|
| 1              | Houston          |
| 4              | Stafford         |
| 5              | Bellaire         |
| 5              | Sugarland        |
| 5              | Houston          |



# SPECIFY REFERENTIAL INTEGRITY CONSTRAINTS

**EMPLOYEE**

|       |       |       |            |       |         |     |        |          |     |
|-------|-------|-------|------------|-------|---------|-----|--------|----------|-----|
| FNAME | MINIT | LNAME | <u>SSN</u> | BDATE | ADDRESS | SEX | SALARY | SUPERSSN | DNO |
|-------|-------|-------|------------|-------|---------|-----|--------|----------|-----|

**DEPARTMENT**

|       |                |        |              |
|-------|----------------|--------|--------------|
| DNAME | <u>DNUMBER</u> | MGRSSN | MGRSTARTDATE |
|-------|----------------|--------|--------------|

**DEPT\_LOCATIONS**

|                |                  |
|----------------|------------------|
| <u>DNUMBER</u> | <u>DLOCATION</u> |
|----------------|------------------|

**PROJECT**

|       |                |           |      |
|-------|----------------|-----------|------|
| PNAME | <u>PNUMBER</u> | PLOCATION | DNUM |
|-------|----------------|-----------|------|

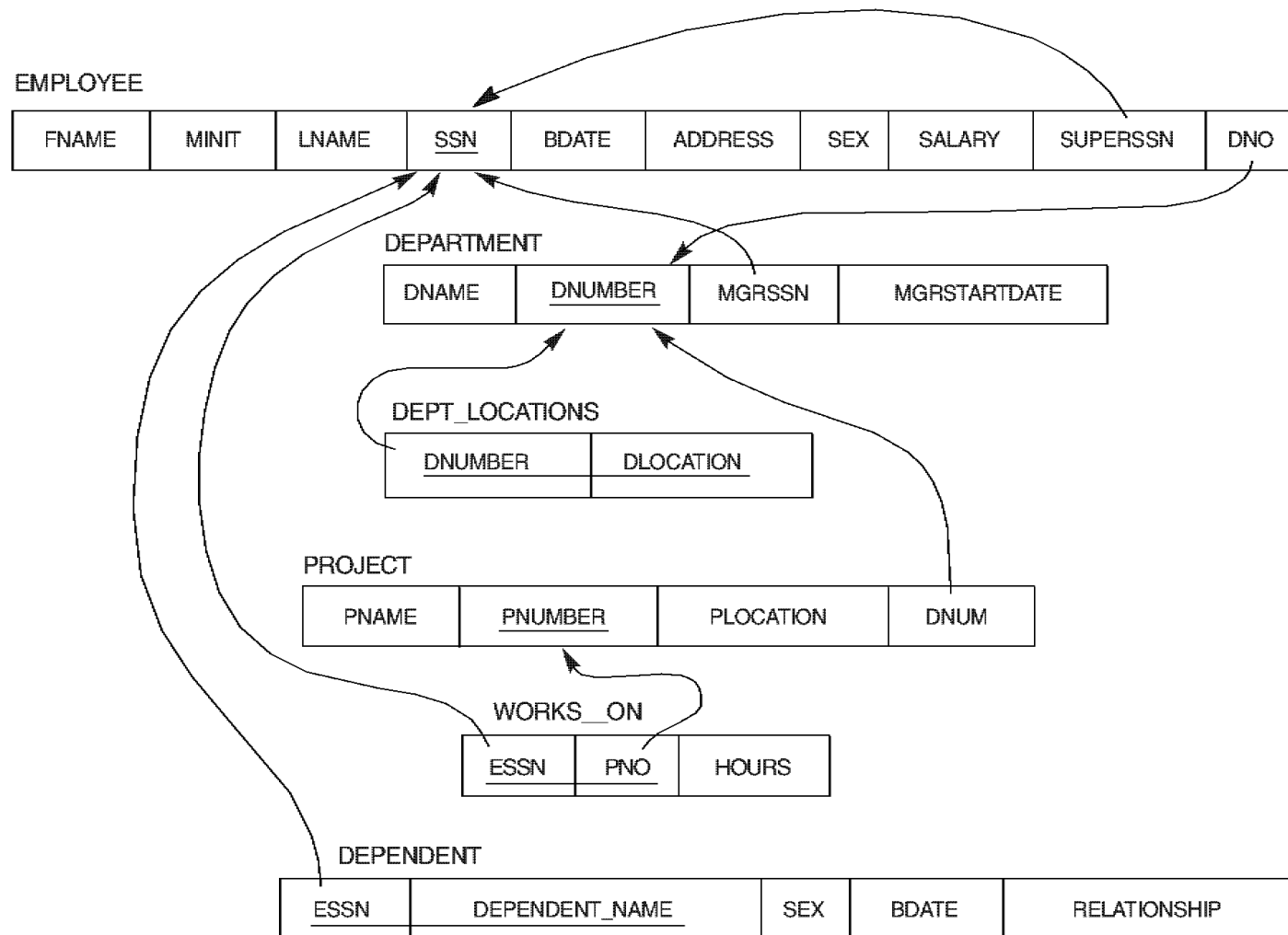
**WORKS\_ON**

|             |            |       |
|-------------|------------|-------|
| <u>ESSN</u> | <u>PNO</u> | HOURS |
|-------------|------------|-------|

**DEPENDENT**

|             |                       |     |       |              |
|-------------|-----------------------|-----|-------|--------------|
| <u>ESSN</u> | <u>DEPENDENT_NAME</u> | SEX | BDATE | RELATIONSHIP |
|-------------|-----------------------|-----|-------|--------------|

**Figure 5.7** Referential integrity constraints displayed on the COMPANY relational database schema diagram.



# FOREIGN KEY (FK)

- A foreign key can be a primary key or any of the candidate key in the referenced relation.
- FK requires that uniqueness constraint should hold for the column on which it is defined in referenced relation.
- *Sql server allows specifying FK on a column which has uniqueness constraint set.*



# SEMANTIC INTEGRITY CONSTRAINTS

- Based on application semantics and cannot be expressed by the model schema
- **Example**
  - The max no of hours per employee for all projects he or she works on is 56 hrs per week
  - Employee salary should not exceed supervisor salary
- SQL-99 allows triggers and **ASSERTIONS** to allow for some of these



# OPERATIONS IN RDBMS

Operations in RDBMS are classified as

- Retrieval
- Update
- **Update Operations on Relations**
  - INSERT a tuple.
  - DELETE a tuple.
  - MODIFY a tuple.
  - Integrity constraints should not be violated by the update operations.



# INSERT OPERATION

- **Insert Operation** can violate following constraints
  - Domain constraint
  - Key constraint (if key value already exists)
  - Entity (if PK contains NULL)
  - Referential (FK refers to a tuple that does not exist in referenced relation)

**EMPLOYEE**

|       |       |       |            |       |         |     |        |          |     |
|-------|-------|-------|------------|-------|---------|-----|--------|----------|-----|
| FNAME | MINIT | LNAME | <u>SSN</u> | BDATE | ADDRESS | SEX | SALARY | SUPERSSN | DNO |
|-------|-------|-------|------------|-------|---------|-----|--------|----------|-----|



# INSERT OPERATION

- Insert <'Cecilia', 'F', 'Kolonsky', NULL, '1960-04-05', '6357 Windy Lane, Katy, TX', F, 28000, NULL, 4> into EMPLOYEE.

EMPLOYEE

| Fname    | Minit | Lname   | <u>Ssn</u> | Bdate      | Address                  | Sex | Salary | Super_ssn | Dno |
|----------|-------|---------|------------|------------|--------------------------|-----|--------|-----------|-----|
| John     | B     | Smith   | 123456789  | 1965-01-09 | 731 Fondren, Houston, TX | M   | 30000  | 333445555 | 5   |
| Franklin | T     | Wong    | 333445555  | 1955-12-08 | 638 Voss, Houston, TX    | M   | 40000  | 888665555 | 5   |
| Alicia   | J     | Zelaya  | 999887777  | 1968-01-19 | 3321 Castle, Spring, TX  | F   | 25000  | 987654321 | 4   |
| Jennifer | S     | Wallace | 987654321  | 1941-06-20 | 291 Berry, Bellaire, TX  | F   | 43000  | 888665555 | 4   |
| Ramesh   | K     | Narayan | 666884444  | 1962-09-15 | 975 Fire Oak, Humble, TX | M   | 38000  | 333445555 | 5   |
| Joyce    | A     | English | 453453453  | 1972-07-31 | 5631 Rice, Houston, TX   | F   | 25000  | 333445555 | 5   |
| Ahmad    | V     | Jabbar  | 987987987  | 1969-03-29 | 980 Dallas, Houston, TX  | M   | 25000  | 987654321 | 4   |
| James    | E     | Borg    | 888665555  | 1937-11-10 | 450 Stone, Houston, TX   | M   | 55000  | NULL      | 1   |

DEPARTMENT

| Dname          | <u>Dnumber</u> | Mgr_ssn   | Mgr_start_date |
|----------------|----------------|-----------|----------------|
| Research       | 5              | 333445555 | 1988-05-22     |
| Administration | 4              | 987654321 | 1995-01-01     |
| Headquarters   | 1              | 888665555 | 1981-06-19     |

DEPT\_LOCATIONS

| <u>Dnumber</u> | <u>Dlocation</u> |
|----------------|------------------|
| 1              | Houston          |
| 4              | Stafford         |
| 5              | Bellaire         |
| 5              | Sugarland        |
| 5              | Houston          |



# INSERT OPERATION

- Insert <'Cecilia', 'F', 'Kolonsky', '677678989', '1960-04-05', '6357 Windswept, Katy, TX', F, 28000, '987654321', 7> into EMPLOYEE.

EMPLOYEE

| Fname    | Minit | Lname   | <u>Ssn</u> | Bdate      | Address                  | Sex | Salary | Super_ssn | Dno |
|----------|-------|---------|------------|------------|--------------------------|-----|--------|-----------|-----|
| John     | B     | Smith   | 123456789  | 1965-01-09 | 731 Fondren, Houston, TX | M   | 30000  | 333445555 | 5   |
| Franklin | T     | Wong    | 333445555  | 1955-12-08 | 638 Voss, Houston, TX    | M   | 40000  | 888665555 | 5   |
| Alicia   | J     | Zelaya  | 999887777  | 1968-01-19 | 3321 Castle, Spring, TX  | F   | 25000  | 987654321 | 4   |
| Jennifer | S     | Wallace | 987654321  | 1941-06-20 | 291 Berry, Bellaire, TX  | F   | 43000  | 888665555 | 4   |
| Ramesh   | K     | Narayan | 666884444  | 1962-09-15 | 975 Fire Oak, Humble, TX | M   | 38000  | 333445555 | 5   |
| Joyce    | A     | English | 453453453  | 1972-07-31 | 5631 Rice, Houston, TX   | F   | 25000  | 333445555 | 5   |
| Ahmad    | V     | Jabbar  | 987987987  | 1969-03-29 | 980 Dallas, Houston, TX  | M   | 25000  | 987654321 | 4   |
| James    | E     | Borg    | 888665555  | 1937-11-10 | 450 Stone, Houston, TX   | M   | 55000  | NULL      | 1   |

DEPARTMENT

| Dname          | <u>Dnumber</u> | Mgr_ssn   | Mgr_start_date |
|----------------|----------------|-----------|----------------|
| Research       | 5              | 333445555 | 1988-05-22     |
| Administration | 4              | 987654321 | 1995-01-01     |
| Headquarters   | 1              | 888665555 | 1981-06-19     |

DEPT\_LOCATIONS

| <u>Dnumber</u> | <u>Dlocation</u> |
|----------------|------------------|
| 1              | Houston          |
| 4              | Stafford         |
| 5              | Bellaire         |
| 5              | Sugarland        |
| 5              | Houston          |

# INSERT OPERATION

- Insert <'Alicia', 'J', 'Zelaya', '999887777', '1960-04-05', '6357 Windy Lane, Katy,TX', F, 28000, '987654321', 4> into EMPLOYEE.

**EMPLOYEE**

| Fname    | Minit | Lname   | <u>Ssn</u> | Bdate      | Address                  | Sex | Salary | Super_ssn | Dno |
|----------|-------|---------|------------|------------|--------------------------|-----|--------|-----------|-----|
| John     | B     | Smith   | 123456789  | 1965-01-09 | 731 Fondren, Houston, TX | M   | 30000  | 333445555 | 5   |
| Franklin | T     | Wong    | 333445555  | 1955-12-08 | 638 Voss, Houston, TX    | M   | 40000  | 888665555 | 5   |
| Alicia   | J     | Zelaya  | 999887777  | 1968-01-19 | 3321 Castle, Spring, TX  | F   | 25000  | 987654321 | 4   |
| Jennifer | S     | Wallace | 987654321  | 1941-06-20 | 291 Berry, Bellaire, TX  | F   | 43000  | 888665555 | 4   |
| Ramesh   | K     | Narayan | 666884444  | 1962-09-15 | 975 Fire Oak, Humble, TX | M   | 38000  | 333445555 | 5   |
| Joyce    | A     | English | 453453453  | 1972-07-31 | 5631 Rice, Houston, TX   | F   | 25000  | 333445555 | 5   |
| Ahmad    | V     | Jabbar  | 987987987  | 1969-03-29 | 980 Dallas, Houston, TX  | M   | 25000  | 987654321 | 4   |
| James    | E     | Borg    | 888665555  | 1937-11-10 | 450 Stone, Houston, TX   | M   | 55000  | NULL      | 1   |

**DEPARTMENT**

| Dname          | <u>Dnumber</u> | Mgr_ssn   | Mgr_start_date |
|----------------|----------------|-----------|----------------|
| Research       | 5              | 333445555 | 1988-05-22     |
| Administration | 4              | 987654321 | 1995-01-01     |
| Headquarters   | 1              | 888665555 | 1981-06-19     |

**DEPT\_LOCATIONS**

| <u>Dnumber</u> | <u>Dlocation</u> |
|----------------|------------------|
| 1              | Houston          |
| 4              | Stafford         |
| 5              | Bellaire         |
| 5              | Sugarland        |
| 5              | Houston          |



# DELETE OPERATION AND CONSTRAINTS

- **Delete operation** can violate
  - Referential Integrity constraint
- Delete the EMPLOYEE tuple with Ssn = '999887777'.
- Delete the EMPLOYEE tuple with Ssn = '333445555'.



**EMPLOYEE**

| Fname    | Minit | Lname   | Ssn       | Bdate      | Address                  | Sex | Salary | Super_ssn | Dno |
|----------|-------|---------|-----------|------------|--------------------------|-----|--------|-----------|-----|
| John     | B     | Smith   | 123456789 | 1965-01-09 | 731 Fondren, Houston, TX | M   | 30000  | 333445555 | 5   |
| Franklin | T     | Wong    | 333445555 | 1955-12-08 | 638 Voss, Houston, TX    | M   | 40000  | 888665555 | 5   |
| Alicia   | J     | Zelaya  | 999887777 | 1968-01-19 | 3321 Castle, Spring, TX  | F   | 25000  | 987654321 | 4   |
| Jennifer | S     | Wallace | 987654321 | 1941-06-20 | 291 Berry, Bellaire, TX  | F   | 43000  | 888665555 | 4   |
| Ramesh   | K     | Narayan | 666884444 | 1962-09-15 | 975 Fire Oak, Humble, TX | M   | 38000  | 333445555 | 5   |
| Joyce    | A     | English | 453453453 | 1972-07-31 | 5631 Rice, Houston, TX   | F   | 25000  | 333445555 | 5   |
| Ahmad    | V     | Jabbar  | 987987987 | 1969-03-29 | 980 Dallas, Houston, TX  | M   | 25000  | 987654321 | 4   |
| James    | E     | Borg    | 888665555 | 1937-11-10 | 450 Stone, Houston, TX   | M   | 55000  | NULL      | 1   |

**WORKS\_ON**

| Essn      | Pno | Hours |
|-----------|-----|-------|
| 123456789 | 1   | 32.5  |
| 123456789 | 2   | 7.5   |
| 666884444 | 3   | 40.0  |
| 453453453 | 1   | 20.0  |
| 453453453 | 2   | 20.0  |
| 333445555 | 2   | 10.0  |
| 333445555 | 3   | 10.0  |
| 333445555 | 10  | 10.0  |
| 333445555 | 20  | 10.0  |
| 999887777 | 30  | 30.0  |
| 999887777 | 10  | 10.0  |
| 987987987 | 10  | 35.0  |
| 987987987 | 30  | 5.0   |
| 987654321 | 30  | 20.0  |
| 987654321 | 20  | 15.0  |
| 888665555 | 20  | NULL  |

**DEPARTMENT**

| Dname          | Dnumber | Mgr_ssn   | Mgr_start_date |
|----------------|---------|-----------|----------------|
| Research       | 5       | 333445555 | 1988-05-22     |
| Administration | 4       | 987654321 | 1995-01-01     |
| Headquarters   | 1       | 888665555 | 1981-06-19     |

**DEPENDENT**

| Essn      | Dependent_name | Sex | Bdate      | Relationship |
|-----------|----------------|-----|------------|--------------|
| 333445555 | Alice          | F   | 1986-04-05 | Daughter     |
| 333445555 | Theodore       | M   | 1983-10-25 | Son          |
| 333445555 | Joy            | F   | 1958-05-03 | Spouse       |
| 987654321 | Abner          | M   | 1942-02-28 | Spouse       |
| 123456789 | Michael        | M   | 1988-01-04 | Son          |
| 123456789 | Alice          | F   | 1988-12-30 | Daughter     |
| 123456789 | Elizabeth      | F   | 1967-05-05 | Spouse       |



# UPDATE OPERATION AND CONSTRAINTS

## ○ Update operation

- that involve no PK or FK cause no problem ... only domain constraint are to be check
- Modifying PK is like deleting a tuple and then inserting another



# UPDATE OPERATION AND CONSTRAINTS

- Update the Dno of the EMPLOYEE tuple with Ssn = '999887777' to 7.
- Update the Ssn of the EMPLOYEE tuple with Ssn = '999887777' to '987654321'.



**EMPLOYEE**

| Fname    | Minit | Lname   | Ssn       | Bdate      | Address                  | Sex | Salary | Super_ssn | Dno |
|----------|-------|---------|-----------|------------|--------------------------|-----|--------|-----------|-----|
| John     | B     | Smith   | 123456789 | 1965-01-09 | 731 Fondren, Houston, TX | M   | 30000  | 333445555 | 5   |
| Franklin | T     | Wong    | 333445555 | 1955-12-08 | 638 Voss, Houston, TX    | M   | 40000  | 888665555 | 5   |
| Alicia   | J     | Zelaya  | 999887777 | 1968-01-19 | 3321 Castle, Spring, TX  | F   | 25000  | 987654321 | 4   |
| Jennifer | S     | Wallace | 987654321 | 1941-06-20 | 291 Berry, Bellaire, TX  | F   | 43000  | 888665555 | 4   |
| Ramesh   | K     | Narayan | 666884444 | 1962-09-15 | 975 Fire Oak, Humble, TX | M   | 38000  | 333445555 | 5   |
| Joyce    | A     | English | 453453453 | 1972-07-31 | 5631 Rice, Houston, TX   | F   | 25000  | 333445555 | 5   |
| Ahmad    | V     | Jabbar  | 987987987 | 1969-03-29 | 980 Dallas, Houston, TX  | M   | 25000  | 987654321 | 4   |
| James    | E     | Borg    | 888665555 | 1937-11-10 | 450 Stone, Houston, TX   | M   | 55000  | NULL      | 1   |

**WORKS\_ON**

| Essn      | Pno | Hours |
|-----------|-----|-------|
| 123456789 | 1   | 32.5  |
| 123456789 | 2   | 7.5   |
| 666884444 | 3   | 40.0  |
| 453453453 | 1   | 20.0  |
| 453453453 | 2   | 20.0  |
| 333445555 | 2   | 10.0  |
| 333445555 | 3   | 10.0  |
| 333445555 | 10  | 10.0  |
| 333445555 | 20  | 10.0  |
| 999887777 | 30  | 30.0  |
| 999887777 | 10  | 10.0  |
| 987987987 | 10  | 35.0  |
| 987987987 | 30  | 5.0   |
| 987654321 | 30  | 20.0  |
| 987654321 | 20  | 15.0  |
| 888665555 | 20  | NULL  |

**DEPARTMENT**

| Dname          | Dnumber | Mgr_ssn   | Mgr_start_date |
|----------------|---------|-----------|----------------|
| Research       | 5       | 333445555 | 1988-05-22     |
| Administration | 4       | 987654321 | 1995-01-01     |
| Headquarters   | 1       | 888665555 | 1981-06-19     |

**DEPENDENT**

| Essn      | Dependent_name | Sex | Bdate      | Relationship |
|-----------|----------------|-----|------------|--------------|
| 333445555 | Alice          | F   | 1986-04-05 | Daughter     |
| 333445555 | Theodore       | M   | 1983-10-25 | Son          |
| 333445555 | Joy            | F   | 1958-05-03 | Spouse       |
| 987654321 | Abner          | M   | 1942-02-28 | Spouse       |
| 123456789 | Michael        | M   | 1988-01-04 | Son          |
| 123456789 | Alice          | F   | 1988-12-30 | Daughter     |
| 123456789 | Elizabeth      | F   | 1967-05-05 | Spouse       |



# OPERATIONS AND CONSTRAINTS

- **Insert Operation** can violate following constraints
  - Domain constraint
  - Key constraint (if key value already exists)
  - Entity (if PK contains NULL)
  - Referential (FK refers to a tuple that does not exist in referenced relation)
- **Delete operation** can violate
  - Referential Integrity constraint
- **Update operation**
  - that involve no PK or FK cause no problem ... only domain constraint are to be check
  - Modifying PK is like deleting a tuple and then inserting another





- In case of integrity violation, several actions can be taken:
  - Cancel the operation that causes the violation (REJECT option)
  - Perform the operation but inform the user of the violation
  - Trigger additional updates so the violation is corrected (CASCADE option, SET NULL option)
  - Execute a user-specified error-correction routine



# IN CLASS EXERCISE

Exercise: Consider the following relations for a database that keeps track of student enrollment in courses and the books adopted for each course:

- STUDENT(SSN, Name, Major, Bdate)
- COURSE(Course#, Cname, Dept)
- ENROLL(SSN, Course#, Quarter, Grade)
- BOOK\_ADOPTION(Course#, Quarter, Book\_ISBN)
- TEXT(Book\_ISBN, Book\_Title, Publisher, Author)

**Draw a relational schema diagram specifying the foreign keys for this schema.**



# ASSIGNMENT #1

Due before start of the class

